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Time-dependent self-diffusion by NMR spin-echo

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The relation between the NMR spin-echo attenuation in the magnetic field gradient and the velocity autocorrelations of molecular random migration has been derived by using the comulant expansion theorem for averaging stochastic processes. It may explain certain cases of time-dependent diffusion observed in many systems. In the limit of long correlation times it is identical to Torrey's well-known formula for spin echo attenuation due to self-diffusion. The method is used to derive the expression for spin-echo attenuation of diffusion in restricted regions.

1. Introduction

The nuclear magnetic resonance spin-echo in a magnetic field gradient is a very useful method for the direct measurement of molecular migrations. Torrey [2] has derived the expression for the spin-echo attenuation due to self-diffusion along a time-dependent magnetic field gradient ($G_x = \delta B/\delta x$):

$$\ln \frac{S(t)}{S_0} = \beta(t) = \gamma^2 D \int_0^t \left| \int_0^u G_x(t')\,\mathrm{d}t' \right|^2 \mathrm{d}u\,, \qquad (1)$$

where γ is the giromagnetic ratio of the nuclei and D is the self-diffusion coefficient. The interaction between particles or the interaction with the lattice influences the molecular migration. Therefore, in a short time interval, there is a strong correlation between particle positions. We have shown [4] that the spin-echo attenuation may include the particle–particle correlation $\langle x(t)x(0)\rangle$:

$$\beta(t) = \frac{\gamma^2}{2} \int_0^t \mathrm{d}t_1 \int_0^t \mathrm{d}t_2\, G_x(t_1)\langle x(t_1)x(t_2)\rangle G_x(t_2)\,. \qquad (2)$$

Therefore eq. (1) is only valid for the Einstein–Fick diffusion [3] when the duration of measurement is long compared to the correlation time of migrations. Nowadays the duration of the shortest NMR spin-echo gradient sequence is a few milliseconds, while the correlation times in most systems are below this value. This means that eq. (2) is useful only in some special cases. With the recent development of NMR techniques for medical application that use very fast magnetic-gradient switching, the time of measurement can be shortened to far below a millisecond. This may lead to a new class of NMR methods for the measurement of the random-walk correlation times. In the following we will show the extended treatment of self-diffusion attenuation of the spin-echo. The result clearly explains the relations between NMR quantities measured and microscopic motion. It follows from a calculation using the comulant expansion theorem [5].

We will also show that the spin-echo attenuation of diffusion in a restricted region of any geometry easily follows from eq. (2). The effect

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of restricted self-diffusion on spin-echo is similar to that of anomalous diffusion and it is sometimes difficult to discriminate between the two effects in the experimental data.

2. Spin-echo attenuation and random migration

2.1. Velocity autocorrelation

In NMR spectroscopy molecular migration affects the spin-echo signal when the magnetic field gradient brings about a nonuniform spin precession. In spin response the phase of a particular spin, $\phi_i(t)$, depends upon spin location and therefore it is sensitive to spin migration. The signal is the sum of the spin contributions:

$$S(t) = \left\langle \sum_i S_{0i}\, e^{i\phi_i(t)} \right\rangle . \tag{3}$$

The average $\langle \cdot \rangle$ includes molecular motion. Together with the magnetic field gradient they modulate the spin phase. Thus the spin phase depends upon the magnetic field gradient $\boldsymbol{G}$ and upon the location $\boldsymbol{r}_i$ of a particular spin:

$$\phi(t) = \gamma \int_0^t \boldsymbol{G}_{\text{eff}}(t')[\boldsymbol{r}_i(t') - \boldsymbol{r}_0] . \tag{4}$$

The RF gradient pulse sequence refocuses the spin-echo phase either with two identical gradient pulses intermediate with the π RF pulse or with two gradient pulses of opposite direction. The effective magnetic field gradient G_{eff} describes the effect of both sequences [4]. The 'per partes' integration of eq. (4) gives the expression

$$\phi_i(\tau) = -\int_0^\tau \boldsymbol{F}(t)\boldsymbol{v}_i(t)\, dt \tag{5}$$

where the spin phase depends upon particle velocity $\boldsymbol{v}_i$ if we consider the fact that the spin-echo phase

$$\boldsymbol{F}(t) = \gamma \int_0^t \boldsymbol{G}_{\text{eff}}(t')\, dt' \tag{6}$$

is equal to zero at the time of phase focusing, $t = \tau$. Thus the spin echo signal is

$$S(\tau) = S_0 \left\langle \sum_i e^{-i\int_0^\tau \boldsymbol{F}(t)\boldsymbol{v}_i(t)\, dt} \right\rangle . \tag{7}$$

The self-diffusion is a stochastic Brownian process, and for any stochastic process an average over the exponent develops into an average over the comulant if the comulant expansion theorem [5] is used. The expansion of eq. (7) gives

$$S(\tau) = S_0 \sum_i e^{-\alpha_i(\tau) - \beta_i(\tau) \cdots} \tag{8}$$

with

$$\alpha_i(\tau) = -\int_0^\tau \boldsymbol{F}(t)\langle \boldsymbol{v}_i(t)\rangle\, dt \tag{9}$$

and with

$$\beta_i(\tau) = \tfrac{1}{2}\int_0^\tau dt_1 \int_0^\tau dt_2\, \boldsymbol{F}(t_1)\langle \boldsymbol{v}_i(t_1)\boldsymbol{v}_i(t_2)\rangle \boldsymbol{F}(t_2) . \tag{10}$$

The second and all higher terms of the comulant sum include the correlation between the particle velocities at different times. Because of the stochastic nature of the interaction with many degrees of freedom we can assume that the particle velocity changes in small independent steps. In this case the central-limit theorem allows a Gaussian approximation of the modulation distribution function. This means that one can neglect all higher correlations but the second one. Thus eq. (8) consists of the spin phase modulated by a molecular flow or a drift change, and of the term that attenuates the signal because of random particle migration. In the following we shall focus our attention on the second term of eq. (10). Since it includes the single-particle velocity autocorrelation function, the spin-echo attenuation may provide information not only about the diffusion constant but also about intermolecular interactions. This happens when the time interval of the spin-echo, τ, is of the same order as the characteristic time of the velocity autocorrelations, τ_c. Usually the time of the spin-echo experiment is much longer than the time scale of the correlation times of molecular random walk, $\tau \gg \tau_c$. In this case the velocity autocorrelation function becomes the delta function

$$\langle v_i(t_1)v_i(t_2)\rangle = 2D\delta(t_1 - t_2) . \quad (11)$$

By inserting this expression in eq. (10), we obtain Torrey's well-known result, eq. (1). To go further, we try to express formula (10) in the frequency domain and find the relation between the attenuation and the spectrum of velocity autocorrelation. By inserting the Fourier transform of the autocorrelation function into eq. (10),

$$\langle v_i(t_1)v_i(t_2)\rangle = \frac{1}{\pi}\int_{-\infty}^{\infty} \tilde{D}_i(\omega)\, e^{i\omega(t_1-t_2)}\, d\omega , \quad (12)$$

we get

$$\beta_i(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega,\tau)\tilde{D}_i(\omega)F(-\omega,\tau)\, d\omega . \quad (13)$$

Equation (13) includes the definition of the gradient Fourier transform

$$F(\omega,\tau) = \int_0^{\tau} F(t)\, e^{i\omega t}\, dt . \quad (14)$$

In eq. (13) $\tilde{D}_i(\omega)$ is the tensor representing the spectrum of autocorrelation between the velocity components. The product in eq. (13) with the phase factor F extracts only the diagonal components of the tensor. The diagonal components of the tensor at zero frequency $\tilde{D}_i(0)$ are the self-diffusion constants. Their values are different in the case of anistropy of self-diffusion.

In NMR measurements molecular autocorrelations are usually short compared to the width of the gradient pulses and their interspacing. Only the long-time tail of the velocity autocorrelation might affect the spin-echo attenuation (fig. 1). In the frequency domain this means that only the

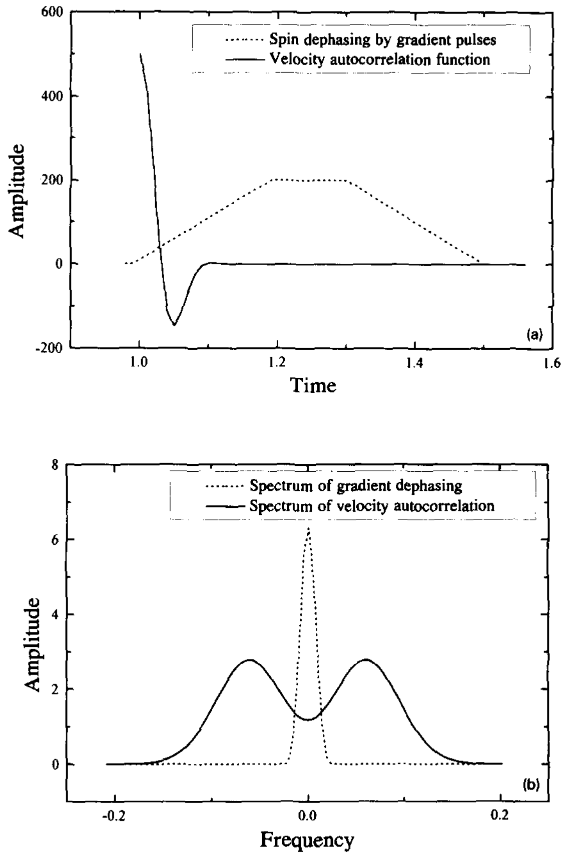

Fig. 1. (a) Spin gradient dephasing and velocity autocorrelation. (b) Fourier transform of the spin dephasing, $F(\omega,t)$, and of the velocity autocorrelation, $D(\omega)$.

spectrum of correlation close to zero frequency contributes to the effect. The spectrum of velocity autocorrelation being a very broad function, eq. (13) takes the form

$$\beta_i(\tau) = \frac{1}{2\pi}\tilde{D}_i(0)\int_{-\infty}^{\infty} |F(\omega,\tau)|^2\, d\omega . \quad (15)$$

Equation (15) becomes identical to Torrey's formula (1) if Parseval's identity is taken into account.

2.2. *Anomalous self-diffusion*

Perhaps the most familiar result derived from Brownian motion theory is Einstein's formula for the averaged square of displacement [3] expressed as a function of time:

$$\langle x^2(t)\rangle = 2Dt\,. \tag{16}$$

D is the diffusion coefficient. Einstein has pointed out that this formula holds only in the limit of large t. Accordingly, as noted by Uhlenbeck and Ornstein [6], several later researchers obtained

$$\langle x^2(t)\rangle = 2D\left[1 - \frac{1}{\beta}(1 - e^{-\beta t})\right] \tag{17}$$

where $\beta = f/m$, f being the coefficient of friction and m the mass of the Brownian particle; this holds for all values of t. Study of the Brownian motion in a periodic potential [7] and in some cases of random walk environments [8] shows a slowing down of diffusion. In several cases the displacement behaves either as

$$\langle x^2(t)\rangle \approx t^a \qquad (a > 1) \tag{18}$$

or as a smooth function of time. This can be explained by the existence of a trapping mechanism, active at very long times. Several measurements of self-diffusion by NMR have shown an anomalous behavior especially in polymers, liquid crystals and other macromolecules [8]. We attribute this to the fact that in these systems the random-walk correlation time becomes comparable to the time of the magnetic-field gradient pulse interspacing. We also believe that the developments in the NMR technique related to its medical applications, such as fast NMR imaging [13], will shorten the time of measurement to less than a millisecond and lead to a new kind of measurement by NMR.

According to eq. (10) the spin-echo attenuation depends upon the convolution between the spectrum of the velocity autocorrelation function and the square of the gradient phase spectrum (fig. 1). This means that the measurement of self-diffusion by NMR might yield information about the velocity autocorrelation and its spectrum. If the time of molecular velocity autocorrelation is not much shorter than the time of the spin-echo measurement, the departure of the spin-echo attenuation from Torrey's approximation can be derived with the use of eq. (10). The long-time tail of the velocity autocorrelation influences the spin-echo; in the frequency domain this means that only the spectrum of correlation close to zero frequency contributes to the effect (fig. 1). The low-frequency expansion of $D(\omega)$ is

$$\tilde{D}_i(\omega) = \tilde{D}_i(0) + \frac{1}{2!}\tilde{D}_i''(0)\omega^2 + \cdots + \frac{1}{2n!}\tilde{D}_i^{(2n)}\omega^{2n}(0) + \cdots\,. \tag{19}$$

By inserting eq. (19) into eq. (13) we obtain

$$\beta_i(\tau) = \frac{1}{2\pi}\sum_{n=0}^{\infty}\frac{1}{2n!}\int_{-\infty}^{\infty}\omega^{2n}\boldsymbol{F}(\omega,\tau)\tilde{D}_i^{(2n)}(0) \times \boldsymbol{F}(-\omega,\tau)\,d\omega \tag{20}$$

$$= \sum_{n=0}^{\infty}\frac{1}{2n!}\int_0^T \boldsymbol{F}^{(2n)}(t)\tilde{D}_i^{(2n)}(0)\boldsymbol{F}^{(2n)}(t)\,dt\,.$$

In Appendix B we have derived $\beta(t)$ for various pulse sequences by using eq. (21).

It is interesting to derive the relation between particle displacement and spin-echo attenuation from eq. (13). The spin echo with infinitely short gradient pulses with width δ and of opposite sign gives

$$\beta_i(\tau) = \frac{1}{2\pi}\gamma^2 G_x^2\delta^2\int_{-\infty}^{\infty} D_{xx,i}\frac{\sin^2(\omega\tau/2)}{(\omega/2)^2}\,d\omega\,, \tag{21}$$

if in eq. (18) $F(\omega,t)$ is substituted by

$$F_x(\omega,\tau) = \gamma G_x\delta\frac{1 - e^{i\omega t}}{i\omega}\,. \tag{22}$$

The definition of the square of the particle displacement is

$$\langle[\boldsymbol{r}_i(t) - \boldsymbol{r}_i(0)]^2\rangle = \int_0^t dt_1\int_0^t dt_2\,\langle\boldsymbol{v}_i(t_1)\boldsymbol{v}_i(t_2)\rangle\,. \tag{23}$$

With the use of eq. (12) it becomes

$$\langle [x_i(t) - x_i(0)]^2 \rangle = \frac{1}{\pi} \int_{-\infty}^{\infty} D_{xx,i}(\omega) \frac{\sin^2(\omega t/2)}{(\omega/2)^2} \, d\omega . \tag{24}$$

By combining eq. (21) and eq. (24) we get

$$\beta_i(\tau) = \tfrac{1}{2}\gamma^2 G^2 \delta^2 \langle [x_i(t) - x_i(0)]^2 \rangle . \tag{25}$$

This is the well-known relation between the spin-echo attenuation and the square of the averaged random-walk displacement. It is a very general relation useful also in the case of anomalous self-diffusion.

3. Restricted self-diffusion

In the following we will show that molecular migration in compartments of a limited size [2] produces effects that can hardly be discriminated from anomalous diffusion in NMR measurements. The diffusion time is the time interval between the first and the second gradient pulse. If it is long enough the reduction of the measured self-diffusion coefficient occurs as a result of molecular collisions with the barriers or a venturing of molecules into regions of decreased mobility. Numerous studies [9] have explored molecular self-diffusion in bounded regions, and a technique has been developed for examining the structural morphology in porous materials since the motion of molecules within a pore is restricted by the surrounding boundaries [10]. We will show how to treat these problems by using a somewhat different approach. The expression describing spin-echo dephasing gives the attenuation related to the correlation of particle positions at different times [4]. We use it to evaluate the effects of restricted diffusion in compartments of different geometry.

If the time of measurement, t, is long with respect to the correlation times, τ_c, the Brownian motion can be considered as a stochastic Markoff process. In evaluating eq. (2) the time averages can be replaced by averages over the ensemble that has the probability functions defined as follows:

(a) $P(\boldsymbol{r}_0,t_1,\boldsymbol{r},t_2)\,d\boldsymbol{r}$ is the probability for the spin which was at the moment t_1 at the position $\boldsymbol{r}_0$, to be at the time t_2 within the volume element $d\boldsymbol{r}$ at the position $\boldsymbol{r}$;

(b) $f(\boldsymbol{r}_0,t)\,d\boldsymbol{r}$ is the probability that the spin is inside the volume $d\boldsymbol{r}$ at the location $\boldsymbol{r}$ at the moment t. Now the correlation function [3] can be written as

$$\boldsymbol{G}_{\text{eff}}(t_1)\langle \boldsymbol{r}(t_1)\boldsymbol{r}(t_2)\rangle \boldsymbol{G}_{\text{eff}}(t_2) - \tag{26}$$

$$\int_V d\boldsymbol{r} \int_V d\boldsymbol{r}_0 \, P(\boldsymbol{r}_0,t_1,\boldsymbol{r},t_2)\boldsymbol{G}_{\text{eff}}(t_1)\boldsymbol{r}_0 \, \boldsymbol{G}_{\text{eff}}(t_2)\boldsymbol{r} \times f(\boldsymbol{r}_0,t_1) . \tag{27}$$

The solution of the problem depends on the construction of the probability functions. If a uniform spin distribution is assumed, $f(\boldsymbol{r}_0,t_1)$ can be replaced by a constant. The function $P(\boldsymbol{r}_0,t_1,\boldsymbol{r}, t_2)$ is the solution of the Einstein–Fick diffusion equation

$$\frac{\delta P}{\delta t} = D \Delta P \tag{28}$$

with the initial condition

$$P(\boldsymbol{r}_0,t_1,\boldsymbol{r},t_1) = \delta(\boldsymbol{r} - \boldsymbol{r}_0) . \tag{29}$$

The general solution for a bounded medium has the form

$$P(\boldsymbol{r}_0,t_1,\boldsymbol{r},t_2) = \sum_k u_k(\boldsymbol{r}_0) u_k(\boldsymbol{r}) \, e^{-a_k D |t_2 - t_1|} \tag{30}$$

where the compartment geometry defines the set of orthogonal functions $u_k(\boldsymbol{r})$. If the gradient points in the direction defined by the unit vector $\boldsymbol{n}$, then $\boldsymbol{G}_{\text{eff}}(t) = g(t)\boldsymbol{n}$. With the substitution of eq. (27) into eq. (2) the spin-echo attenuation becomes

$$\beta(\tau) = \tfrac{1}{2} \sum_k B_k \int_0^T dt_1 \int_0^T dt_2 \, g(t_1) g(t_2) \, e^{-a_k D |t_1 - t_2|} \tag{31}$$

where

$$B_k = \int_V \mathrm{d}\boldsymbol{r} \int_V \mathrm{d}\boldsymbol{r}_0 \, \boldsymbol{nr}\, \boldsymbol{nr}_0 \, u_k(\boldsymbol{r}) u_k(\boldsymbol{r}_0) \,. \tag{32}$$

If the geometry of the compartments is known the coefficients B_k and a_k can be determined. Appendix B shows the solutions for the planar, cylindrical and spherical geometries.

In the case of a pulse sequence that consists of two identical gradient pulses with width δ and separated by τ, formula (31) for spin-echo attenuation becomes

$$\beta(2\tau) = 2\left(\frac{\gamma G}{D}\right)^2 \sum_k \frac{B_k}{a_k^2} \times [a_k D\delta - 1 + \mathrm{e}^{-a_k D\delta} + \mathrm{e}^{-a_k D\tau}(1 - \cosh a_k D\delta)] \,. \tag{33}$$

When the time of measurement is short compared to the time needed by the molecule to reach the compartment boundary, $a_k D \gg \tau$, then the above expression becomes the well-known result for diffusion in the infinite medium

$$\beta(2\tau) = \gamma^2 G^2 \delta^2 D\left(\tau - \frac{\delta}{3}\right) \sum_k B_k a_k \tag{34}$$

if the fact that $\Sigma_k B_k a_k = 1$ is taken into account. In the case of a magnetic-field gradient pulse sequence where $\delta = \tau$, eq. (31) becomes

$$\beta(2\tau) = \left(\frac{\gamma G}{D}\right)^2 \sum_k \frac{B_k}{a_k^2 D^2} \times (2a_k D\tau - 3 + 4\,\mathrm{e}^{-a_k D\tau} - \mathrm{e}^{-2a_k D\tau}) \,. \tag{35}$$

Anderson [9] obtained the same result (eq. (35)) when he studied self-diffusion through planar layers, using a different approach.

When eq. (31) is transformed into the frequency domain we get the apparent spectrum of the restricted self-diffusion:

$$D_{\mathrm{rest}}(\omega) = \sum_k B_k \frac{a_k D\omega^2}{a_k^2 D^2 + \omega^2} \,. \tag{36}$$

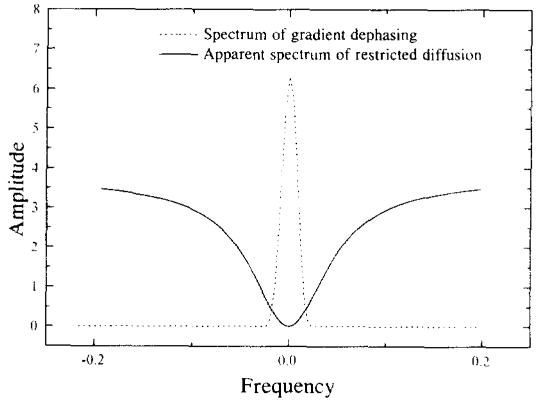

Fig. 2. At low frequencies the apparent spectrum of restricted diffusion may behave like a spectrum of velocity autocorrelation shown on fig. 1.

The shape of this spectrum at low frequencies (fig. 2) is similar to the spectrum of the velocity autocorrelation (fig. 1). The self-diffusion constant in the infinite medium is defined as the autocorrelation spectrum at zero frequency, $D(0)$, while the apparent spectrum of the self-diffusion in a bounded medium, eq. (36), gives $D_{\mathrm{rest}}(0)$ equal to zero. This is obvious since when the measurement time is very long, the self-diffusion coefficient is zero as a result of molecular collisions with the barriers. But when the barriers are permeable and particles venturing into the region of decreased mobility, $D_{\mathrm{rest}}(0)$ may also be different from zero.

For short diffusion times, i.e. $\tau \to 0$, i.e. large frequencies, $\omega \to \infty$, $D_{\mathrm{rest}}(\omega)$ is always equal to the self-diffusion constant D.

We can conclude that the effect of restricted self-diffusion on spin-echo cannot always be clearly discriminated from the effect of strong single-particle autocorrelations [12]. It means that the restricted self-diffusion exhibits a similar long-time and low-frequency dependence as the so-called anomalous self-diffusion.

Appendix A

Attenuation in first approximation

(1) For the usual spin-echo pulse sequence with two identical gradient pulses with width δ

and interspace τ, pointed along the x-axis, the relation of the attenuation (eq. (13)) to the spectrum $D(\omega)$ and the derivatives $D(\omega)^{(2n)}$ at zero frequency is

$$\beta(2\tau) = \gamma^2 G^2 \left\{ D(0)\left[\delta^2\left(\tau - \frac{\delta}{3}\right)\right] + D''(0)\delta + \cdots \right\}. \tag{37}$$

(2) In the case of gradient pulses shaped as a sinus wave with amplitude G the attenuation becomes

$$\beta(2\tau) = \gamma^2 G^2 \times \left[D(0)\left(\frac{2\delta}{\pi}\right)^2\left(\tau - \frac{\delta}{4}\right) + \sum_{n=1}^{\infty} D^{2n}(0)\left(\frac{\pi}{\delta}\right)^{2(n-1)} \delta \right]. \tag{38}$$

(3) In the case of a constant gradient the spin-echo attenuation is

$$\beta(2\tau) = \gamma^2 G^2 \left[D(0)\frac{2\tau^3}{3} + D''(0)\tau + \cdots \right]. \tag{39}$$

Appendix B

B.1. Planar layers

For the planar impermeable layers equally separated by the distance d, the solution of eq. (27) is

$$P(x_0,t_1,x,t_2) = \frac{2}{d}\left[\frac{1}{2} + \sum_{n=1}^{\infty} \cos\frac{n\pi x_0}{d} \cos\frac{n\pi x}{d}\, e^{-(n\pi/d)^2 D|t_1-t_2|}\right] \tag{40}$$

with the gradients applied perpendicularly to the layers. Hence, it is possible to determine the constants

$$B_k = \frac{8d^2}{(2k-1)^4\pi^4}, \tag{41}$$

$$a_k = \frac{\pi^2(2k-1)^2}{d^2}. \tag{42}$$

B.2. Cylindrical boundaries

For a self-diffusion in cylindrical tubes with radius R the probability is

$$P(r_0,\varphi_0,t_1,r,\varphi,t_2) = \sum_{n,k} \frac{\varepsilon_n}{\pi R^2(1 - n^2/\mu_k^{(n)2})J_n^2(\mu_k^{(n)})} J_n\left(\frac{\mu_k^{(n)} r_0}{R}\right) \tag{43}$$

$$\times J_n\left(\frac{\mu_k^{(n)} r}{R}\right) \cos n\varphi_0 \cos n\varphi\, e^{-(\mu_k^{(n)}/R)^2 D|t_1-t_2|}. \tag{44}$$

With the gradient perpendicular to the tube axis the correlation function has the form

$$G^2\langle r(0),r(t)\rangle = G^2 \sum_{k=0}^{\infty} \frac{2(R/\mu_k)^2}{\mu_k^2 - 1} e^{-(\mu_k/R)^2 D|t|} \tag{45}$$

with μ_k being the kernels of $J_1'(\mu) = 0$ ($\mu_k = 1.84, 5.33, 8.54, \ldots$),

$$B_k = \frac{2(R/\mu_k)^2}{\mu_k^2 - 1} \tag{46}$$

and

$$a_k = \left(\frac{\mu_k}{R}\right)^2. \tag{47}$$

B.3. Spherical boundaries

Self-diffusion inside a sphere can be evaluated using the probability function

$$P(r_0,\varphi_0,t_1,r,\varphi,t_2) = \sum_{m,n,k} \frac{A_{m,n,k}}{r_0 r} J_{n+1/2}\left(\frac{\mu_m^{(n)} r_0}{R}\right) J_{n+1/2}\left(\frac{\mu_m^{(n)} r}{R}\right) \tag{48}$$

$$\times P_{n,k}(\cos\vartheta_0) P_{n,k}(\cos\vartheta) \cos k\varphi_0 \cos k\varphi \tag{49}$$

$$\times e^{-(\mu_m^{(n)}/R)^2 D|t_1-t_2|} \tag{50}$$

where R is the radius of the spherical compartments and

$$A_{m,n,k} = \varepsilon_k \frac{\pi R^2 (n+k)!}{(2n+1)(n-k)} \left(1 - \frac{n(n+1)}{\mu_m^{(n)2}}\right) J_{n+1/2}(\mu_m^{(n)}) \tag{51}$$

with

$$\varepsilon_k = \begin{cases} 2, & k=0, \\ 0, & k \neq 0. \end{cases} \tag{52}$$

The correlation function (8) is

$$G^2\langle r(0), r(t)\rangle = G^2 \sum_{m=0}^{\infty} \frac{2(R/\mu_m)^2}{\mu_m^2 - 1} e^{-(\mu_m/R)^2 D|t|} \tag{53}$$

with μ_m defined as the kernels of the equation

$$\mu J'_{3/2}(\mu) - \tfrac{1}{2} J_{3/2}(\mu) = 0$$
$$(\mu_m = 2.08, 5.94, 9.20, \ldots).$$

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